Started on	Saturday, 10 October 2026, 10:16 PM
State	Finished
Completed on	Saturday, 10 October 2026, 10:26 PM
Time taken	9 mins 53 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem Statement:

Given a sorted array and a value x, the floor of x is the largest element in array smaller than or equal to x. Write divide and conquer algorithm to find floor of x.

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Value for x

Output Format

First Line Contains Integer – Floor value for x

Answer: (penalty regime: 0 %)

```
#include<stdio.h>

int floorSearch(int a[], int low, int high,int x){
    if(low>high)
      return -1;
    int mid = low+(high-low)/2;
    if(a[mid]==x)
      return a[mid];
     if(a[mid]>x)
      return floorSearch(a, low, mid -1,x);
     int r = floorSearch(a, mid+1, high,x);
     return(r!=-1)?r:a[mid];

}

int main(){
    int n,x;
    scanf("%d",&n);
    int a[n];
    for(int i = 0;i<n;i++)
        scanf("%d",&a[i]);
    scanf("%d",&x);


    printf("%d\n",floorSearch(a,0,n-1,x));
    return 0;

}
```

	Input	Expected	Got	
✔	6 1 2 8 10 12 19 5	2	2	✔
✔	5 10 22 85 108 129 100	85	85	✔

	Input	Expected	Got	
✔	7 3 5 7 9 11 13 15 10	9	9	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.